THE STRUCTURE OF MATHEMATICAL REALITY

Phase III — Chapter 8

Vector Spaces and the Meaning of "Canvas"

Quiz

Self-Study Curriculum

Before you begin

This quiz tests application, not memorisation. Answer in your own words.

Do not quote the lecture back at yourself. If you find yourself writing a phrase you remember reading, stop and rephrase it from your own understanding. A short, precise answer in your own language is worth more than a long transcription.

1. Two data scientists analyze the same dataset. One uses raw features; the other uses PCA-transformed features. They run the same algorithm and get different results. Who is "right"? What structural assumption distinguishes their analyses?

2. The function spaces used in quantum mechanics are infinite-dimensional vector spaces. What does it mean for a space to be "infinite-dimensional"? What is the basis, and what plays the role of coordinates?

3. You rotate a dataset before clustering it. The clusters change. What does this tell you about the relationship between the clustering algorithm and the inner product?

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4. A nonlinear system can sometimes be "linearized" around an equilibrium point. In the language of this chapter, what is being approximated, and what structural assumption is being made?

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After the quiz

For each answer, ask yourself: *"Did I explain the **why**, or did I just state the **what**?"* If you stated a fact without explanation, go back and add the reasoning.

Date: ____________________ **Questions where I want to go deeper:** _________________________